

STUDENT PERFORMANCE ANALYSIS

Portfolio Data Analysis Report — Simulated Dataset

Executive Summary: This project analyzes a simulated dataset of 600 students. The analysis examines pass/fail outcomes, subject averages, attendance, study hours and their relationships with final performance. The overall simulated pass rate is 100.00%. Attendance and study hours are positively associated with final score in this generated dataset.

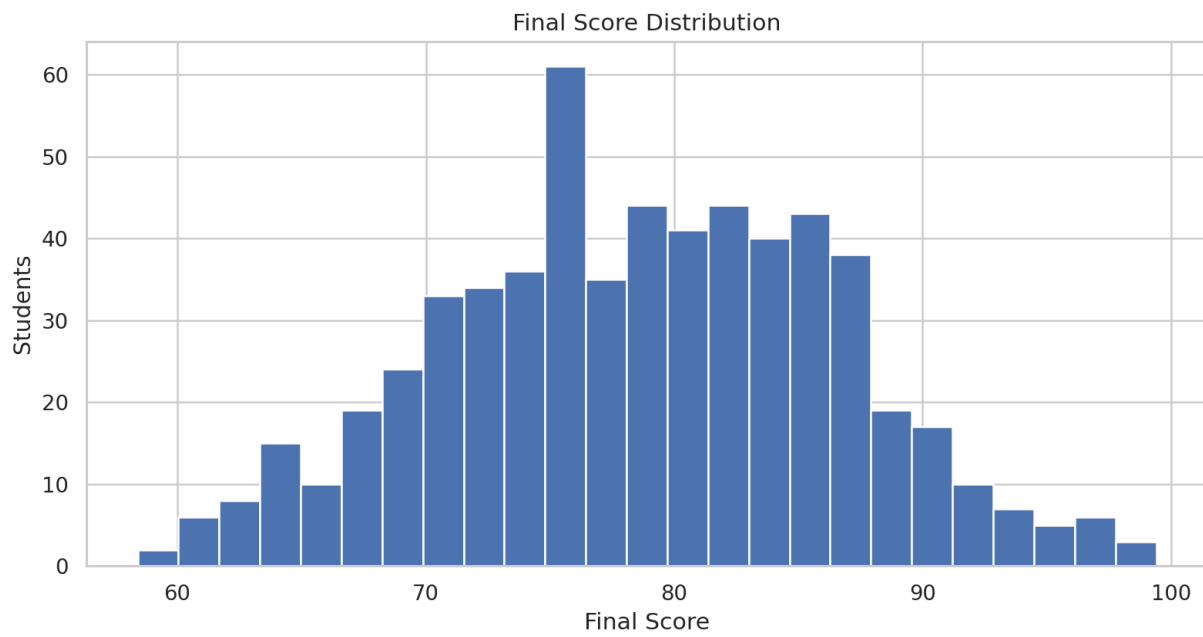
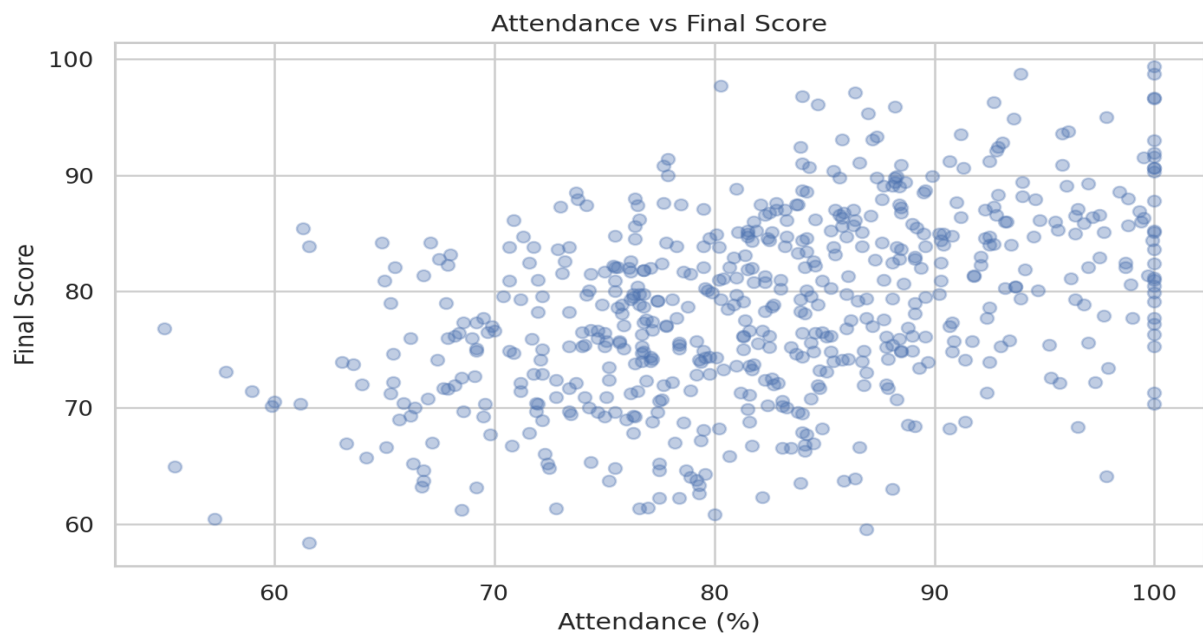
1. Data Quality

The dataset contains 600 rows and 10 columns. Validation confirms no missing cells, no duplicate rows, and valid 0–100 ranges for attendance and scores.

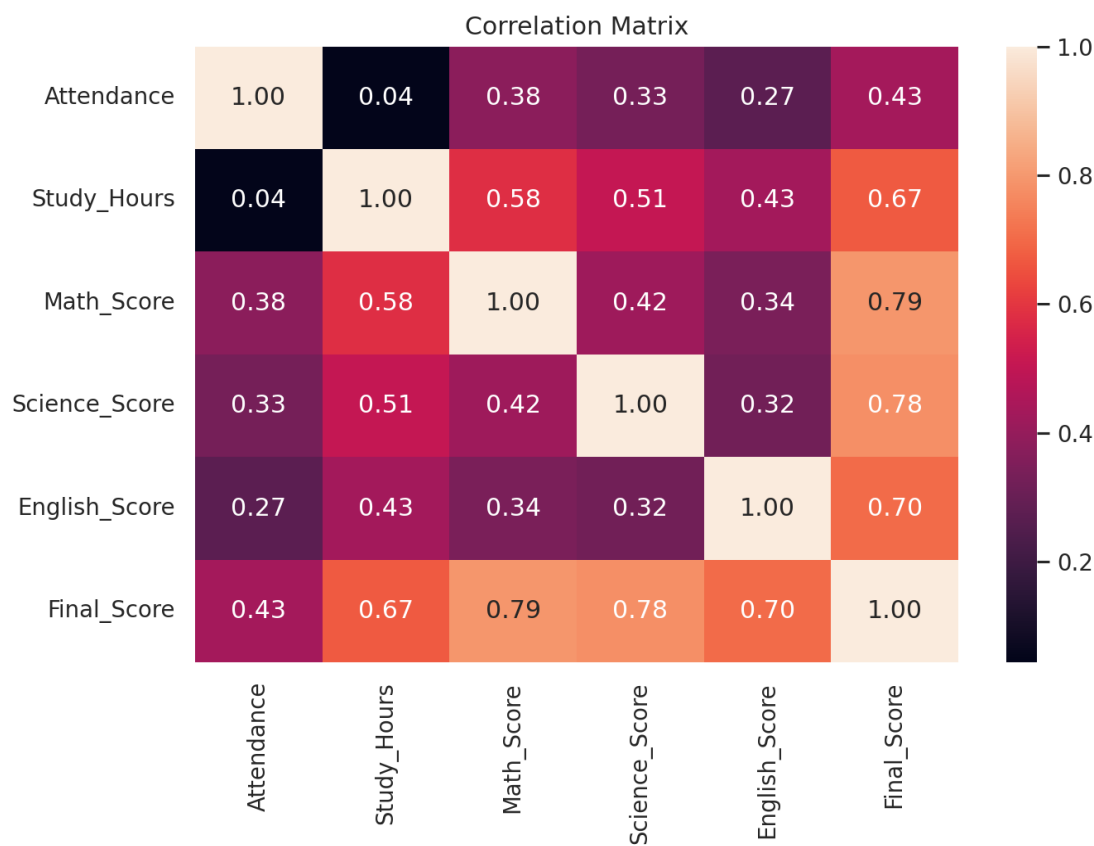
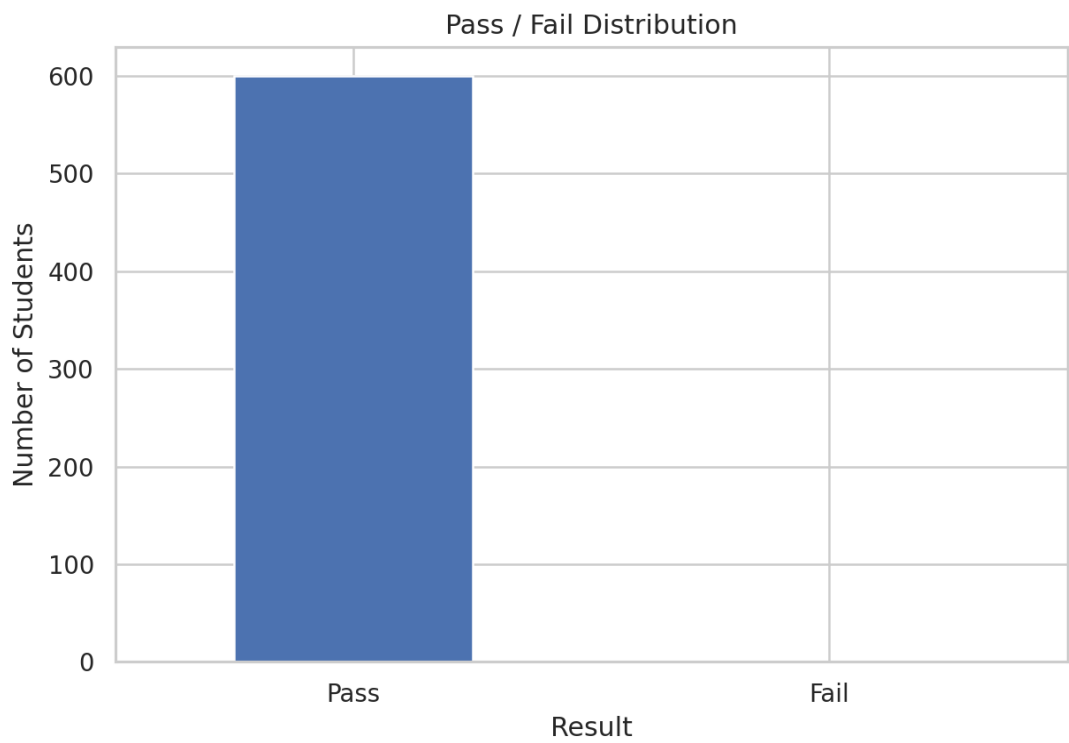
Metric	Value
Students	600
Pass rate	100.00%
Fail rate	0.00%
Attendance / Final correlation	0.433
Study hours / Final correlation	0.671
High attendance mean	82.53
Low attendance mean	72.91
Welch t-test p-value	5.85176e-18

2. Visual Analysis





3. Additional Analysis



4. Key Findings

- The simulated pass rate is 100.00%.
- Attendance and final score correlation is 0.433.
- Study hours and final score correlation is 0.671.
- Students with attendance $\geq 85\%$ average 82.53, compared with 72.91 for students below 70% attendance.
- The Welch t-test p-value is 5.85176e-18; interpretation should use a preselected significance level and recognize that the data are simulated.

5. Recommendations

- Use attendance as one component of an early-warning framework.
- Provide targeted academic support to students showing both low attendance and low scores.
- Compare subject-level outcomes to identify areas for additional support.
- Monitor study habits together with academic outcomes.
- Validate findings with real institutional data before making operational decisions.

6. Limitation

This dataset is simulated and is intended only for demonstrating data-analysis techniques. The findings must not be represented as evidence about a real student population.